



Document Effective Date
20/09/2023

DEVELOPING PROCESS EVENT SCENARIOS GUIDELINE

DOC NO: IS-HSE-GDL-310

Rev: 01

Quality Reviewed By: OIB/4

 Digitally signed
by Moklis Miah
Date: 2023.09.20
06:55:36 +03'00'

CUSTODIAN DEPARTMENT	DOCUMENT AUTHOR	DOCUMENT OWNER
HEALTH, SAFETY & ENVIRONMENT (IDD EL-SHARGI)	Bothaina Al-abdulla  Digitally signed by Bothaina Al-abdulla Date: 2023.09.19 12:52:27 +03'00'	Tigor Maruli  Digitally signed by Tigor Maruli Date: 2023.09.19 13:20:46 +03'00'
	OIH/41	A/OIH/4

FINAL APPROVAL
 IBRAHIM AL-AHMAD
OIH

	DEVELOPING PROCESS EVENT SCENARIOS GUIDELINE DOC NO: IS-HSE-GDL-310 REV. 01
---	--

TABLE OF CONTENTS

	Page No.
1.0 PURPOSE	3
2.0 SCOPE AND APPLICABILITY.....	3
3.0 GUIDELINE	3
3.1 CHOOSING AN EVENT SCENARIO	3
3.2 STEPS IN DEVELOPING AN EVENT SCENARIO	6
3.3 CREDIBLE & NON-CREDIBLE EVENT SCENARIOS	12
4.0 REFERENCES TO OTHER DOCUMENTS	15
5.0 DEFINITIONS.....	16
APPENDIX.....	17
APPENDIX A - EVENT TREE FOR ESTABLISHING EVENT SCENARIOS.....	17
APPENDIX B - LEVELS OF CAUSALITY	19
APPENDIX C -TYPICAL OFFSHORE HAZARD SCENARIOS	22
APPENDIX D - REVISION HISTORY LOG	24
APPENDIX E - LIST OF ABBREVIATIONS	24

	DEVELOPING PROCESS EVENT SCENARIOS GUIDELINE DOC NO: IS-HSE-GDL-310 REV. 01
---	--

1.0 PURPOSE

This Guideline is intended to provide a system for the revision of Process Hazard Review (PHR) Documents / reports.

The General objectives of the Idd El Shargi Risk Management Guidelines (RMG) within the Idd El Shargi RM process are to:

- Allow a clearer definition of requirements with regard to the Idd El Shargi RM program and related policies and Procedures;
- Allow specific topics to be addressed in detail relevant to the application of the RM element, including examples where relevant, and
- Act as a basis for further information, understanding and training.

Note: These Documents should be considered as Guidelines only and not mandatory unless stated. For further clarification on content and intent contact Idd El Shargi HSE Risk Engineer.

The intent of this specific Guideline over and above the General Requirements is to ensure that, when evaluating Risk, only credible Event Scenarios are developed and also to promote consistency in the development of Event Scenarios upon which risk assessments are conducted. It also provides guidance on the recommended steps that should be taken to develop an Event Scenario.

2.0 SCOPE AND APPLICABILITY

This Guideline applies to Employees, contractors, and subcontractors, whether engaged directly by Idd El Shargi or by others, who will perform work or services of any kind at any Idd El Shargi facilities and/or locations.

The Guideline applies to all personnel within Idd El Shargi plants and facilities (PS-1, ISND, ISSD and ISHT), who are required to develop Event Scenarios, using IS-HSE-PRC-301, the Risk Evaluation and Response Procedure.

3.0 GUIDELINE

3.1 CHOOSING AN EVENT SCENARIO

Development of credible Event Scenarios is a critical first step in evaluating Risk. The Risk Assessment Team must be familiar with the process and the associated Hazards. An Event Scenario is not the worst case scenario but rather a realistic event that has a reasonable likelihood to result in harm to people, the environment or Company assets or earnings.

An Event Scenario consists of initiating / escalating event(s) leading to exposure to an identified Hazard, and an estimated Consequence of exposure. For each identified Hazard, an Event Scenario shall be developed if there are any of the following credible Consequences:

- a. Human Injury/Illness at a Facility
- b. Human Injury Outside a Facility
- c. Facility Property Damage and Loss of Revenue
- d. Non-Owned Property Damage

	DEVELOPING PROCESS EVENT SCENARIOS GUIDELINE DOC NO: IS-HSE-GDL-310	REV. 01
---	--	----------------

- e. Environmental Impairment: Clean-up / Remediation Cost
- f. Environmental Impairment: Habitat or Wildlife Population

Where there is no credible potential for the above Consequences there is no credible Event Scenario. The proposed Event Scenario is deemed not credible and, therefore, Risk assessment is unnecessary. Figure 1 below depicts the progression of an Event Scenario.

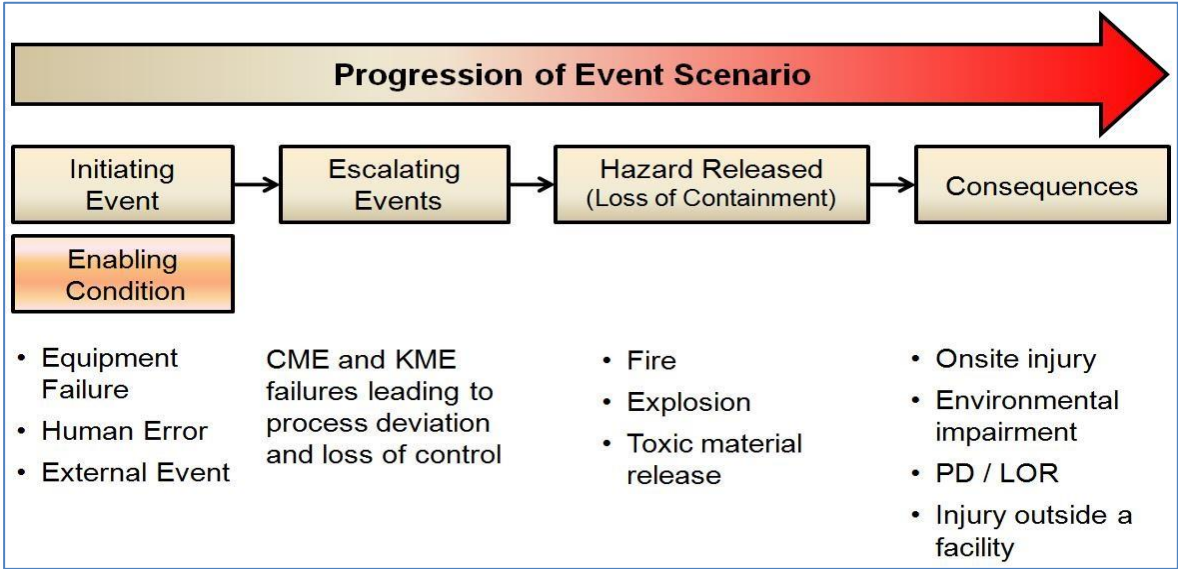


FIGURE 1. EVENT SCENARIO PROGRESSION

As shown in Figure 1, an initiating event such as an equipment failure or human error begins the sequence of events that leads to Consequences. For many Process Event Scenarios¹, multiple Critical Mitigating Elements (CMEs) should function to prevent escalation of events. If the Key Mitigating Elements (KMEs), that support the CMEs are not appropriately maintained, the CMEs will tend to degrade over time, resulting in lower CME reliability or effectiveness. Under these conditions, an Event Scenario can progress and allow loss of control over a Hazard, typically in the form of loss of containment of a hydrocarbon or toxic material. After a Hazard is “released”, there should be other CMEs available that are designed to prevent or mitigate Consequences of the Hazard. However, if these CMEs are not effective, then the Event Scenario may progress until damage or harm occurs, resulting in an ultimate undesired Consequence. In some cases, if an Event Scenario is not stopped from progressing, then Knock-on Effects may occur whereby the Consequences initiate another Event Scenario(s) having more severe or different Consequences (domino effect). Therefore, it is desirable to place priority on CMEs and supporting KMEs to prevent the Hazard from being released.

For some Event Scenarios, an enabling condition shall exist concurrent with the initiating event in order for the scenario to commence. For example, a pump failure that occurs during the warm climate season may have no adverse Consequences. However, if it fails during cold ambient temperature conditions, then the product may solidify in the piping causing a rupture and adverse Consequences. Note that if this enabling condition of freezing

¹ Non-Process Event Scenarios are similar except that Hazards are not process equipment-related and harm typically results from impacts, collisions, or falls rather than loss of containment.

	DEVELOPING PROCESS EVENT SCENARIOS GUIDELINE DOC NO: IS-HSE-GDL-310 REV. 01
---	--

temperature exists for less than 10% of the year, then a CME Contingent Element, may be credited to recognize the low likelihood of this occurrence.

Table 1 shows some common terms used to describe components of Event Scenarios that are utilized by different risk assessment methodologies.

TABLE 1. EVENT SCENARIO TERMS

General Term:	Initiating Event	Safeguard to Prevent or Mitigate Loss of Control of Hazard	Loss of Control Over a Hazard	Safeguard to Prevent or Mitigate Consequences	Safeguard to Maintain Another Safeguard
Critical Mitigation Elements Analysis:	Initiating Event	CME (strong safeguard) ² KME (weak safeguard)	Loss of Containment (typical)	CME (strong safeguard) ² KME (weak safeguard)	KME
HAZOP:	Cause	Safeguard (strong or weak)	Immediate Consequence	Safeguard (strong or weak)	Safeguard
LOPA:	Initiating Event	Independent Protection Layer (IPL)	Immediate Consequence	Independent Protection Layer (strong)	Safeguard (non- IPL)
Bow-Tie:	Threat	Proactive Control or Threat Barrier	Top Event	Reactive Control or Recovery Measure	Escalation Factor Control

² Only CMEs may be used for probability reduction credits in a PHR

	DEVELOPING PROCESS EVENT SCENARIOS GUIDELINE DOC NO: IS-HSE-GDL-310 REV. 01
---	--

3.1.1 Source of Information to Assist in Developing Event Scenarios

The Risk Assessment team shall review the following:

- a. Any prior incidents from Idd El Shargi and others
- b. Facility siting studies
- c. Lessons learned
- d. Prior Process Hazard Reviews and Risk Assessment
- e. Past experience of longer-term employees and Contractors.

3.1.2 Developing an Event Scenario

A Process Event Scenario typically consists of an initiating event and a subsequent series of events resulting in loss of containment (release scenario). After a release scenario is developed, additional series of events / situations are identified that result in one, or more, Consequences.

3.2 STEPS IN DEVELOPING AN EVENT SCENARIO

The sequence of steps for developing Event Scenarios will vary depending upon the selected method. For example, in a HAZOP study, after a process deviation is selected the first step will be to identify initiating events (causes) for a postulated Event Scenario. The team will then analyse the sequence of events that could lead to loss of control over a Hazard. In contrast, a Bow-Tie analysis will commence with identification of Hazards, followed by a description of the manner in which there is loss of control over the Hazard. The following sections cover the key steps associated with the progression of Event Scenarios.

3.2.1 Identify Hazards

Determine the Hazards associated with the process or activity being evaluated. A hazard is defined as a characteristic, activity, arrangement, event, process or substance that has the potential to adversely affect (i.e., cause damage or harm) to people, the environment, property or revenue. Process Hazards could include flammability, toxicity, high/low temperature, high pressure, reactivity, electrical systems, etc. Non-process hazards could include operations and activities such as ground transportation in mountainous terrain and desert environments, aviation transportation, marine transportation and heavy equipment activities (e.g., drilling and well servicing operations).

It is important to consider factors that can give rise to Hazards such as location, phase of operation or activity, and abnormal conditions. As an example, diversion of flow into a different tank or system to conduct a maintenance activity may result in an H₂S Hazard that would not normally be present. An example of a non-process Hazard involving ground transportation is driving downhill during freezing weather conditions.

3.2.2 Identify Initiating Events

An initiating event is the first event or failure, usually in a series of events that can lead to loss of control over a Hazard. These are listed in the "Cause" column of a HAZOP worksheet. In this step the team attempts to identify all possible causes that could lead to release of a Hazard, ignoring any CMEs, that may be in place to prevent escalation of the event to loss of control over the Hazard.

Initiating events can be described by various levels of causes, ranging from an immediate cause (for a process deviation) to a root cause that points to a management system failure.

	DEVELOPING PROCESS EVENT SCENARIOS GUIDELINE DOC NO: IS-HSE-GDL-310 REV. 01
---	--

An appropriate level of cause for Event Scenarios is considered a “basic cause” as it provides more detail for risk assessment and aids in deciding what risk reduction measures may be needed. As shown in Appendix 3 – Levels of Causality, an immediate cause for high level in a tank could be a pump fails, ‘off’. However, this level is too general to identify what actions that can prevent or mitigate the pump from failing, ‘off’. By iteratively asking “why” the failure occurred, more specific causes, including root causes, can be identified similar to that used in some incident investigation methods. For the purpose of developing Event Scenarios and subsequent risk assessment, it is neither practical nor efficient to evaluate failures to a root cause level. In general, Process Hazard Review (PHR) teams should identify basic causes and, in some instances, may need to evaluate more specific causes.

An initiating event is not itself the Hazard. For example, a fire is a thermal radiation Hazard to persons or equipment, but it is not considered an initiating event. Therefore, a HAZOP worksheet “Cause” column should not list a fire, explosion, or toxic release as the initiating event. Rather, the initiating event would typically be an equipment failure or human error which could lead to a loss of containment for which there could be fire, explosion, or toxic Hazards. An exception might be an external event involving one of these Hazards for which Idd El Shargi has no control and would not generally be able to identify a specific cause. For example, a toxic release from a neighbouring facility could be considered an initiating event assuming this was deemed a credible scenario.

The major categories of initiating events are:

- Equipment failures
- Human failures / errors
- External events.

The following sections provide examples of these types of initiating events:

a. Equipment Failures

Failures of equipment are among the most common initiating events that can lead to adverse consequences. Typical equipment failures are:

- Controls failure (consider malfunction of a loop as a whole, rather than discrete components, causing actuation of the final control element).
- Rotating equipment (e.g., compressor, fan, or pump) failure, rotating equipment driver failure (e.g., motor or engine), or rotating equipment coupling failure.
- Corrosion, erosion, and cracking failure mechanisms of piping and equipment.
- Valve leak-by, failing open or closed.
- Incorrect design rating or inappropriate materials leading to internal or external corrosion / erosion / stress cracking (could be considered human error).

b. Human Failures / Errors

This category includes errors involving any persons interacting with Processes such as operators, maintenance technicians, and inspection and testing service personnel. These errors can involve acts of omission or commission, meaning a required action was not

	DEVELOPING PROCESS EVENT SCENARIOS GUIDELINE DOC NO: IS-HSE-GDL-310 REV. 01
---	--

performed, was performed incorrectly, or additional actions were performed. Examples include:

- Inadvertent operation of an in-service valve, valve left open or closed following maintenance.
- Startup/shutdown of wrong equipment.
- Adding wrong material.
- Omitting or adding an operating step.
- Incorrect sequence or timing of operating steps.
- Incorrect software or firmware programming of a control system or instrumented protection system.
- Incorrect calibration or ranging of instrumentation.
- Vehicular impacts with unguarded equipment.
- Mis-operation of equipment, dropping an object onto equipment.

Inadvertent operation of an in-service open-ended drain or vent valve (for which any release would be obvious), should not normally be considered an initiating event unless it is a quarter-turn valve for which the handle can be accidentally moved. However, a drain / vent valve left open following maintenance or leak-by of a drain / vent valve could be considered credible initiating events. CMEs Car-Sealed or Locked Valve, and Bull Plug Program, might apply in these cases.

Incorrect programming of control systems or instrumented protection systems can be an initiating event for process deviations. This would not normally be considered in a typical PHR but would be considered an initiating cause in a specific controls system review.

c. External Events

External events can be initiating events for process deviations. Examples are:

- Physical damage from severe weather (high winds / knock-on effects, precipitation, flooding).
- Electrical power loss causing equipment shutdown or loss of another utility system.
- High ambient temperature or solar radiation causing hydraulic (thermal) expansion / pressure build-up (high ambient temperature is a cause for high pressure).
- Low ambient temperature causing increased fluid viscosity / flow reduction.
- Low ambient temperature causing fluid freezing or plugging, increased fluid viscosity, flow reduction, and vacuum collapse conditions.
- Contractor excavation striking an underground pipeline.

Before considering environmental / weather-related events as initiating causes, the facility design should be evaluated to determine if such events are credible. For example, if a facility is designed for the maximum wind speed in that region, then high wind would not be considered a credible cause for physical damage unless the facility has degraded to a level such that structural integrity is questionable.

d. Abnormal Conditions

	DEVELOPING PROCESS EVENT SCENARIOS GUIDELINE DOC NO: IS-HSE-GDL-310 REV. 01
---	--

Consideration should also be given to abnormal conditions involving any of the above categories of initiating events or that are enabling conditions for Event Scenarios such as:

- Partial electric power failure allowing continued operation of some equipment while other equipment or protective systems are impaired / inoperable, creating abnormal operating conditions.
- Steam drying equipment that can cause vacuum conditions / collapse when temperature of shut-in equipment drops.
- Opening process equipment exposed to hydrocarbon streams with H₂S that can result in iron sulfide scale (pyrophoric) ignition with oxygen.
- Tank draining when vacuum protection has been removed, causing lower pressure / collapse.
- Various causes of blocked centrifugal pump discharge (no flow) resulting in high temperature fluid vaporization and high pressure.

3.2.3 Identify Escalating Events Leading to Release of Hazard

Escalating events are the subsequent series of events that could occur following an initiating event in the absence of any CMEs to stop the progression of events. In this step the team should assume that all CMEs fail to function as designed even if they are very robust.

Some examples of escalating events are:

- Level in separator drops, allowing gas blow-by to downstream KO drum; pressure rises in KO drum and exceeds equipment design limits, causing KO drum to rupture.
- Pressure rises in V-101 and equalizes; inability to transfer liquid out causes high level in V-101 and liquid carryover to KO drum; high level in KO drum and carryover into compressor leading to compressor failure.
- Release of flammable gas from corroded pipe (In this example, the initiating event may be normal corrosion or another mechanism such as failure of corrosion inhibitor system that results in accelerated corrosion).
- Dropped object impacts 1" pipe.
- Unfavorable weather conditions leads to high seas and poor visibility; inability of marine vessel thrusters to maintain control; vessel strikes HP gas riser.

3.2.4 Describe Release of Hazard

The potential for Consequences exists when there is loss of control over a Hazard, allowing it to be "released". For Processes, this typically involves loss of containment of process materials, though it could also involve mechanical loss of containment. In this step, the team should describe the loss of control over the Hazard and provide a relative scale of the Hazard being released. Examples are:

- Large release of hydrocarbon from rupture of pressure vessel under 500 psig.
- Medium-size release of hydrocarbon through 1-inch hole in equipment at 50 psig.
- Gasket leak on 3,000 psig hydrocarbon piping.
- Release of toxic gas from PSV opening to atmosphere (200 lb/min).
- 5 kW turbine over-speed failure.

	DEVELOPING PROCESS EVENT SCENARIOS GUIDELINE DOC NO: IS-HSE-GDL-310 REV. 01
---	--

- Crude oil overflows storage tank at 20 gpm.

For a case of overpressure (i.e., system/process pressure that exceeds the design pressure) a loss of containment scenario must be considered. However, consideration should be given to factors that may influence the size of the release such as:

- Rate of pressure increase (slow pressure increase may manifest in a pipe/flange leak rather than vessel rupture).
- Rate of depressurization.
- Weakest pressure-containing component of system failing first.

For Non-Process Event Scenarios, a Hazard is typically released in the form of a collision, impact, exposure, or fall. Examples of these mechanisms are:

- Material transport truck roll-over while going downhill.
- Light vehicle collision with another light vehicle.
- Person falling to ground from structure.
- Helicopter holding up to 14 persons crashes into water.

3.2.5 Identify Escalating Events after Release of Hazard

Following release of the Hazard, the team should identify the subsequent series of events that could occur, if any, again assuming the absence of any CMEs to stop the progression of events. In many cases, such as an explosion, there are no escalating events except for possible Knock-on effects.

Some examples of escalating events are:

- Released hydrocarbon gas drifts into and fills an adjacent processing unit and ignites.
- Crude oil from overflowing storage tank flows down adjacent ditch and into stream leading to bay.

3.2.6 Determine the Consequences of the Event

Consequences can be developed once a Hazard has been released and the escalating events, if any, are identified. It is important to evaluate Consequences under the assumption that CMEs and KMEs that may be in place to prevent escalation of the event do not function as intended. This approach should then lead to the credible worst case Consequences for the Event Scenario. Typically, there is an intermediate Consequence from a release of a hydrocarbon – a fire, explosion or both. The team should be cognizant of possible Knock-on Effects that can occur from the intermediate Consequence. For example, a jet fire from a flange leak that ignites might impinge upon an adjacent condensate line, causing it to rupture and ignite in a larger fire. Typically, there is no intermediate Consequence for a toxic release because the release directly impacts people. The final Consequences can be damage or harm in the form of injury to people, environmental impairment, damage to company assets/loss of production revenue or damage to third-party property. The Idd El Shargi HSE Risk Matrix defines seven Consequence categories though, at most, only six of these categories are applicable to Process Risks (Human Injury Outside a Facility Related to Security or Transportation Scenarios would not apply to process events). Each Consequence category should be evaluated separately as to applicability for an Event

	DEVELOPING PROCESS EVENT SCENARIOS GUIDELINE DOC NO: IS-HSE-GDL-310 REV. 01
---	--

Scenario. For each Consequence category that applies the Event Scenario development team needs to estimate the Consequence level.

The following Event Scenario example has several Consequences:

- a. **Initiating event:** An operator opens a bypass around a pressure control valve to a low-pressure vessel which is not designed to handle the upstream pressure.
- b. **Escalating event leading to Hazard:** The pressure in the vessel increases above the design limits.
 - Assume that the high-pressure alarm / operator action (CME) is ineffective.
 - Assume that the independent high-pressure shutdown (CME) is ineffective.
 - Assume that the pressure relief device (CME) is ineffective.
- c. Release of Hazard (loss of containment): Vessel ruptures
- d. Escalating event after release of hazard: Flammable or toxic material fills the process unit.
 - Assume that appropriate electrical classification (CME) is ineffective.
- e. **Intermediate Consequences of the event:**
 - The flammable material ignites resulting in a fire/explosion.
- f. **Final Consequences of the event:**
 - Human Injury at a Facility:
 - Applicable - An employee is exposed to a high concentration of H₂S which would likely result in a Permanent Incapacitating Injury (PII)
 - Facility Property Damage and Loss of Revenue (excluding Clean-up/Remediation):
 - Applicable - The vessel is damaged beyond repair requiring replacement and resulting in a 1-month loss of production
 - Impacts Outside a Facility (except Security or Transportation Scenarios):
 - Not Applicable
 - Impacts Outside a Facility Related to Security or Transportation:
 - Not Applicable
 - Damage to Property Not Owned or Operated by Idd El Shargi or a Contractor (excluding Clean-up/Remediation):
 - Not Applicable
 - Environmental Impairment – Clean-up/Remediation Cost:
 - Applicable – Clean up and disposal of charred debris for USD \$50,000
 - Environmental Impairment – Habitat or Wildlife Population:
 - Not Applicable.

3.2.7 Event Scenario Exceptions

A manned location is defined as having persons assigned on a permanent basis where the Event Scenario is expected to impact a person.

	DEVELOPING PROCESS EVENT SCENARIOS GUIDELINE DOC NO: IS-HSE-GDL-310 REV. 01
---	--

If a location is not manned, an on-site injury or PII should not be considered credible. The Event Scenario would still need to be evaluated for an off-site injury or PII and for any other potential Consequences. OSHA considers a facility to be unmanned if employees are at the site for an average of less than 1.5 man-hours per day and less than a total of 14.5 man-hours per week.

3.3 CREDIBLE & NON-CREDIBLE EVENT SCENARIOS

An undesired Event Scenario is one that is considered credible because it has a reasonable probability of resulting in one or more Consequences. “Reasonable probability” means that most people with the appropriate knowledge and experience would consider the Event Scenario possible.

An Event Scenario with a Probability of $<10^{-6}$ per year in quantitative terms would be assigned a Probability level of “1” per the Idd El Shargi risk matrix which implies that it is still credible at this low likelihood. To qualitatively estimate such a low frequency is difficult as a person’s experience level to gauge Probability is typically 10 - 40 years. Therefore, when deciding whether or not an Event Scenario should be considered credible, the guidance provided here is that persons evaluating risk should be highly certain that the event will not happen, to consider it as not being credible.

Using the CME approach for probability estimation, a single CME that is capable of preventing an Event Scenario must not be assumed to make the scenario non-credible. An Event Scenario might be considered as non-credible, for example, if the Event Scenario is overpressure and there is no source of pressure that exceeds the design pressure of the system (accounting for current MI condition of the system). However, experienced judgment and supporting data should be applied to ensure that it is reasonable to consider a scenario as credible or non-credible, accounting for factors such as reliability of human performance, historical failure frequencies of equipment, and the possibility of degraded CMEs.

The following sections provide guidance for considerations that may apply when determining if an Event Scenario is credible or non-credible.

3.3.1 Designs to Prevent Event Scenarios

A fundamental assumption made in PHRs is that when process parameters cannot exceed the design limits of the system or equipment under study, then there is no process deviation and, hence, no credible Event Scenario. This assumption is predicated on the following factors for existing systems and equipment:

- Designed to industry standards and codes (to provide suitable design factor).
- Constructed under appropriate standards and quality assurance.
- Operated within the design parameters.
- Maintained with appropriate inspection, testing, maintenance (ITM) programs.

If there are no sources of pressure that could exceed the design pressure of a vessel and the vessel has been appropriately designed, constructed, operated, and maintained, then an overpressure Event Scenario could be considered non-credible. An overpressure (or other parameter deviation) Event Scenario should be considered credible if there is not a high degree of confidence that the above factors are well supported as is often the case for older facilities in which documentation or history may be difficult to obtain and verify. If the above factors can be substantially verified but there is question as to the reliability of the factors or unknown factors such as a possible but unlikely corrosion mechanism, then the

	DEVELOPING PROCESS EVENT SCENARIOS GUIDELINE DOC NO: IS-HSE-GDL-310 REV. 01
---	--

scenario should be considered credible. However, a CME Innately Safe Design, might still apply.

When planned facilities are being analysed such as in a detailed design phase PHR, consideration should be given to the expected robustness of future construction operation, and maintenance activities when considering an Event Scenario non-credible. If there is supporting evidence that these factors will be addressed for the planned facilities, then it may be appropriate to consider a postulated Event Scenario “non- credible”.

3.3.2 Double / Multiple Jeopardy

Double / multiple jeopardy is sometimes used in PHRs as a basis for declaring a postulated scenario not credible (i.e., non-credible). The following criteria should be used by risk evaluators to determine if double / multiple jeopardy applies:

- The postulated scenario requires two or more unrelated events or conditions to occur concurrently or be present in order to initiate the sequence of events that could lead to adverse Consequences.
- The events or conditions should both have a very low likelihood of occurrence such that the combined probability is so low as to consider the Event Scenario “not credible”.
- There are no common-cause failure mechanisms between the events or conditions.

Double / multiple jeopardy is NOT an occurrence of:

- An initiating event and failure of a robust CME that prevents escalation of an Event Scenario.
- Failure of two or more CMEs.

These types of failures may have a low probability of occurrence, yet are still credible and so would not be considered double / multiple jeopardy.

Double / multiple jeopardy is similar to a postulated scenario that requires an enabling condition to initiate the sequence of events except that the enabling condition would have to be present for only a very short time in order for the probability to be so low as to consider the Event Scenario not credible. An example of a postulated Event Scenario in which double jeopardy might be claimed as a reason to consider a scenario non-credible is:

A LIGHTNING STRIKE (INITIATING EVENT) IGNITES A COLD VENT WHEN GAS IS BEING DISCHARGED TO THE VENT DUE TO A PROCESS UPSET (INFREQUENT SECOND EVENT OR ENABLING CONDITION), RESULTING IN A FIRE THAT DAMAGES PROXIMATE OVERHEAD CABLES.

For considering this Event Scenario not credible, it would need to be confirmed that vent is normally isolated therefore gas discharges to the vent are very uncommon (e.g., several hours per year or less) and that the facility is in a location that is not prone to frequent lightning strikes. Further, if it is reasonable that a lightning strike could trigger a facility shutdown, thereby causing gas to vent, then this postulated scenario would be considered credible (common-cause failure mechanism for presence of gas and ignition of the gas). In

	DEVELOPING PROCESS EVENT SCENARIOS GUIDELINE DOC NO: IS-HSE-GDL-310 REV. 01
---	--

this example, if there are no adverse Consequences to a fire at the cold vent, then postulated Event Scenario would neither be considered credible nor assessed for risk.

3.3.3 Unmanned Locations

A manned location is defined as having persons assigned on a permanent basis where the Event Scenario is expected to impact a person. If a location is not manned, an on- site injury or PII would not be considered credible. The premise for this assumption is similar to double jeopardy in which two or more unrelated events or conditions having low probability of occurrence would have to occur concurrently for a PII to occur. To consider an Event Scenario non-credible for an unmanned location, the likelihood of an event that could cause injury should be very low. (Of course, the Event Scenario would still need to be evaluated for an off-site injury or PII and for other potential Consequences).

OSHA considers a facility unmanned if employees are at the site for an average of less than 1.5 man-hours per day and less than a total of 14.5 man-hours per week. This can be used as a guide to determine if a facility is manned and injury to people should be evaluated as a credible consequence.

An example of a permanent manned location would be a Gas Plant with onsite operators. A location is not considered manned if a person infrequently visits a location within a field or plant (e.g., during field or plant rounds).

Some examples of Scenarios where Consequences to people would not be credible for an Event Scenario:

When onsite consequences to people should not be considered:	A production tech visits a tank battery to gauge tanks. They spend about an hour on site each day (Days only - 7 hours per week). A production tech uses a dog house located near a central tank battery to fill out paperwork - they spend about an hour each day (Days only - 7 hours per week). A plant operator making rounds in a compressor building 4 times per shift and spends about 15 minutes in the building (Plants operate 24/7 – 14 hours per week). The potential consequence is very localized such that a person would have to be immediately adjacent to the event to be impacted – e.g., a small-bore H₂S release in any area where people are not normally present and escape is not obstructed.
When onsite consequences to people should be considered:	Control Room located at a manned gas plant, central battery or compressor station. A production area uses an office located near a central tank battery (or compressor station) as a meeting point to issue permits and jobs and fill out paperwork - occupied about three hours each day (5 days only - 15 hours per week).

	DEVELOPING PROCESS EVENT SCENARIOS GUIDELINE DOC NO: IS-HSE-GDL-310 REV. 01
---	--

4.0 REFERENCES TO OTHER DOCUMENTS

TABLE 1: REFERENCES TO OTHER DOCUMENTS

Document Title	Document Number	Internal/External Document (if any)	Document Referencing
Risk Evaluation & Response Procedure	IS-HSE-PRC-301	Internal	Upwards
Critical Mitigation Elements (CME) Methodology Guideline	IS-HSE-GDL-309	Internal	Sideways

	DEVELOPING PROCESS EVENT SCENARIOS GUIDELINE DOC NO: IS-HSE-GDL-310 REV. 01
---	--

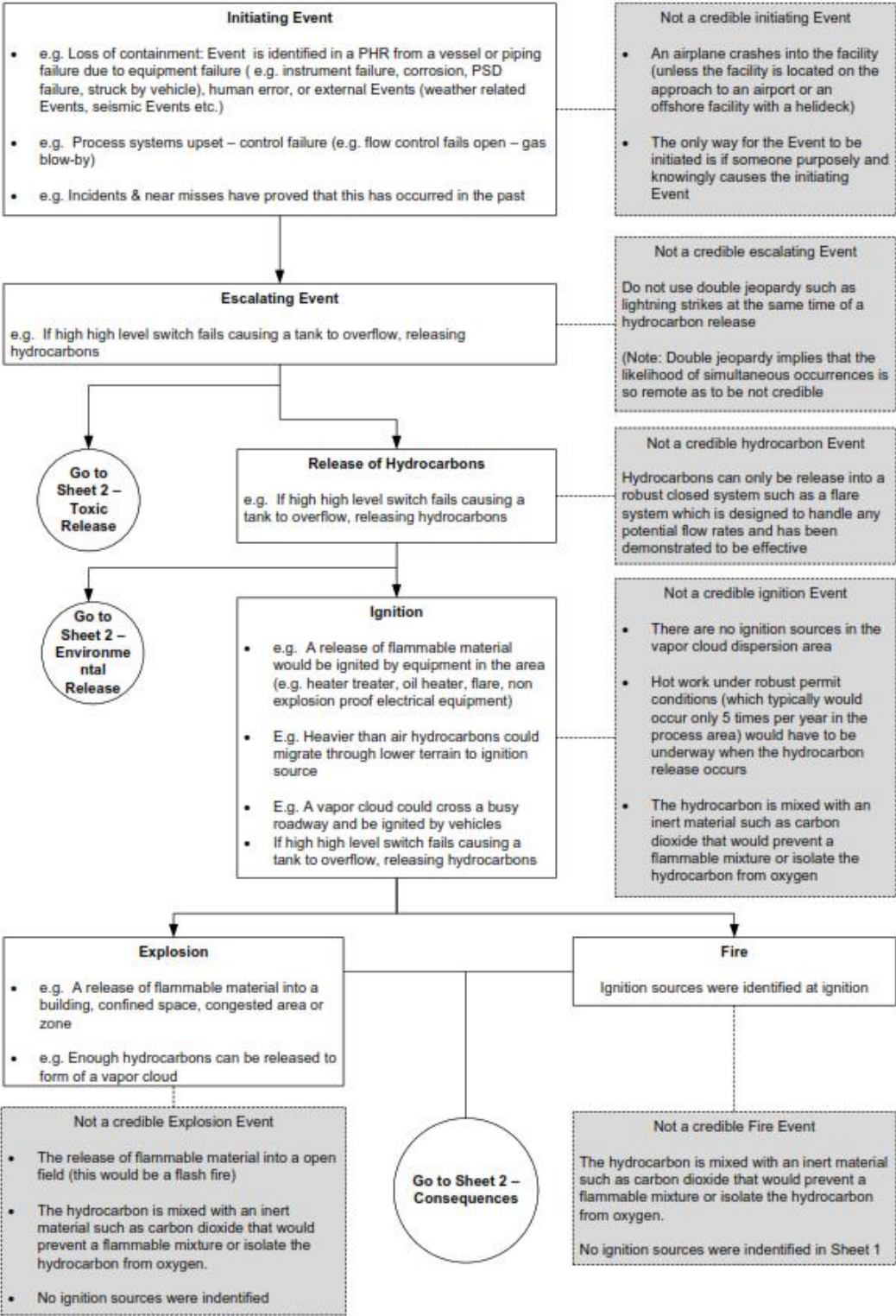
5.0 DEFINITIONS

TABLE 2: KEY TERMS DEFINED

Term	Definition
CME	Critical Mitigation Element, An independent and reliable device or system that is capable of: 1) preventing initiation of an Event Scenario, 2) preventing loss of Hazardous Material containment, or 3) substantially mitigating one of an Event Scenario's Consequences. Failure of one CME shall not prevent another CME from performing its function. CME shall be maintained to the reliability level assumed in the risk analysis. A CME reduces an Event Scenario's Probability by one level on the Risk Matrix.
HAZOP	Hazard and Operability Study A formal team-based, structured, hazard identification and risk assessment method that applies a detailed deviation, guideword, parameter, cause, consequence, and controls approach. Its main objective being to validate a robust design in terms of Hazards And Operability- it is not intended as a design review.
KME	Key Mitigation Element HSE Risk mitigation elements (equipment, design features, specifications, inspection and maintenance routines, and operating parameters) that are necessary to achieve and maintain the reliability level of Critical Mitigation Elements (CMEs), and Risk Mitigation Measures (RMM), or risk mitigations that are not otherwise claimed as CMEs / RMMs in a risk analysis.
PII	Permanent Incapacitating Injury A fatality or physical injury or illness that permanently renders a person incapable of performing the essential duties of any of the occupations for which a person is qualified by education, training and experience
PHR	Process Hazard Review A systematic approach for identifying and assessing Process Hazards and risks and their possible effects.

APPENDIX

APPENDIX A - EVENT TREE FOR ESTABLISHING EVENT SCENARIOS

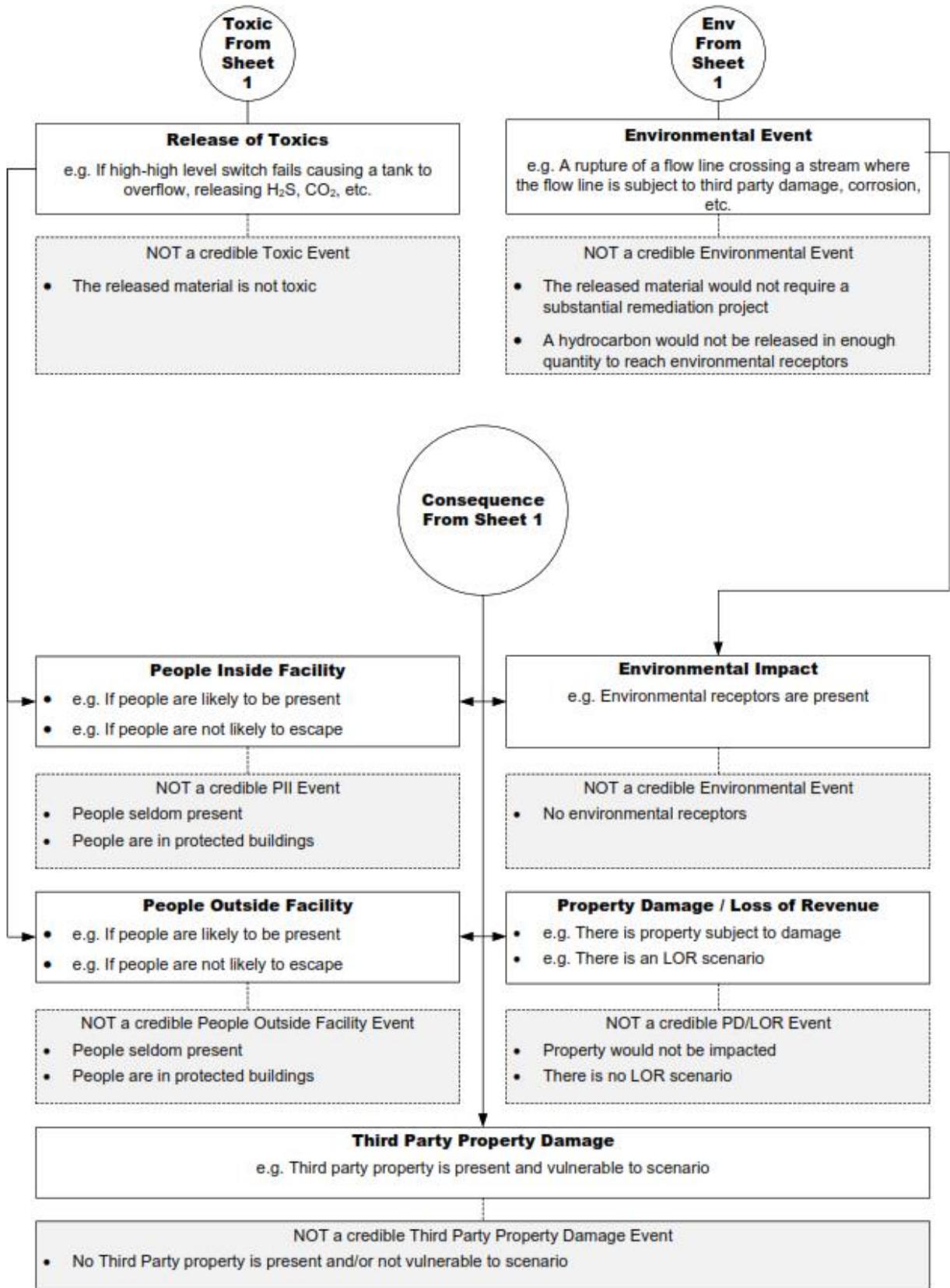




DEVELOPING PROCESS EVENT SCENARIOS GUIDELINE

DOC NO: IS-HSE-GDL-310

REV. 01



	DEVELOPING PROCESS EVENT SCENARIOS GUIDELINE DOC NO: IS-HSE-GDL-310 REV. 01
---	--

APPENDIX B - LEVELS OF CAUSALITY

For the purposes of developing Event Scenarios in PHRs, initiating events will normally be described by Level 2 – Basic Causes. Depending upon the scenario, enabling causes or even root causes may need to be evaluated for recommendations to be made that address the initiating event.

ITM – Inspection, Testing, Maintenance

LTA – Less Than Adequate

Level 1 Immediate Cause	Level 2 Basic Causes	Level 3 Enabling Causes	Level 4 Root Causes
<i>Why?</i>	<i>Why?</i>	<i>Why?</i>	<i>Why?</i>
Pump P-101A/B fails, 'off' (resulting in high level and overflow from a tank)	1. Mechanical failure		
	bearing failure	Lack of or inadequate ITM Poor pump rebuild practices	Ineffective PM program No PM program LTA training LTA skill LTA performance
	shaft break	Cavitation	LTA process design Plugged strainer
		Minimum flow region	Impeller too large & throttled
	debris jams in impeller	Lack of strainer Debris in process fluid	LTA maintenance practices LTA process design
	coupling break	Wrong size or type	LTA mechanical design
		Flow surge Improper alignment	LTA process design LTA training LTA skill LTA performance

	DEVELOPING PROCESS EVENT SCENARIOS GUIDELINE DOC NO: IS-HSE-GDL-310 REV. 01
---	--

	motor failure	Lack of or inadequate ITM End of life.	LTA training LTA skill LTA performance
	2. Pump switched off (human action)	Act of omission or commission	LTA training LTA skill LTA performance LTA communication
	3. Loss of electrical power to equipment hardware failure utility failure	Lack of or inadequate ITM Lack of or inadequate ITM LTA electrical power design	LTA training LTA skill LTA performance (may not be within ability to control)
Initiating Factor	Hazard Group	Event-Consequence	
Corrosion Defect / fatigue Earthquake Extreme weather Scour Subsidence	Structural Failure	Platform collapse Local damage Containment failure	
Corrosion Defect / fatigue Earthquake Operator error	Equipment Failure	Containment failure Projectiles Dropped Object	
Equipment failure Operator error	Dropped object	Containment failure Equipment failure	
Corrosion Defect / fatigue Erosion Impact Operator error Structural failure	Containment Failure	Fire, Explosion, Radiation, Smoke Asphyxiation Poison Energy release Environmental damage	
Corrosion Defect / fatigue	Helicopter incident	Containment failure Structural failure	

	DEVELOPING PROCESS EVENT SCENARIOS GUIDELINE DOC NO: IS-HSE-GDL-310 REV. 01
---	--

Extreme weather Operator error		
Corrosion Defect Operator error	Control system / safety System failure	Containment failure
Defect Fire Operator error	Non-Process fire or explosion	Projectiles Smoke / Fumes
Defect Extreme weather Operator error	Vessel collision	Structural failure Containment failure
Containment failure	Fire / explosion	Toxic gas / asphyxiation
Defect Extreme weather Obstruction Operator error	Occupational incident	Movement inhibition

	DEVELOPING PROCESS EVENT SCENARIOS GUIDELINE DOC NO: IS-HSE-GDL-310 REV. 01
---	--

APPENDIX C -TYPICAL OFFSHORE HAZARD SCENARIOS

Offshore Hazards – by Type		
PLATFORM ACTIVITIES 1. Working at Height 2. Noise 3. Manual Handling 4. Posture 5. Over side Work 6. Paint Spraying 7. Grit Blasting 8. Helicopter Operations 9. Process Venting 10.N.O.R.M. Scale 11.Turbine Exhausts 12.Deck Drains 13.Release of Hydrocarbons 14.Confined Space / Low oxygen Environment 15.Hazardous Materials 16.Water Jetting 17.HP Grit Blasting 18.High Pressure Hydraulics 19.Rotating / Moving Machinery 20.High Pressure 21.Uncontrolled Pressure Release 22.Mechanical Failure 23.Hot work – Sparks / weld material 24.Restricted Access 25.Thermal condition – heat / cold 26.Close proximity to live electrics 27.Ignition source 28.Radioactivity 29.Toxic / Flammable gas release 30.Dust 31.Fumes 32.Inadequate Lighting Levels		LIFTING OPERATIONS 1. Rigging (SWL/Inspected) 2. Swinging Loads 3. Dropped Loads 4. Heavy Lift 5. Internal Lift 6. Vessel Lift 7. Close proximity to live pipework 8. Weather 9. Crush Potential 10.Subsea assets 11.Skidding Loads
MARINE OPERATIONS 1. Supply Ships 2. Diesel Engine Exhausts 3. Weather 4. Surface Visibility 5. Strong Tidal Flow 6. Alarms – Platform Response 7. Collision 8. Anchor Handling Operations 9. Anchor Patterns (Chain / Wire) 10. Helicopter Operations 11. Alongside Platform 12. DP Failure 13. Sea fastenings 14. Hot Work 15. Lifeboat Launch – DSV Position 16. Alarms – DSV Response		MOBILISATION 1. Transport 2. Cranes 3. Vehicles 4. Forklift Operations 5. Use of Tuggers to Skid Loads 6. Deck Overcrowding 7. Slips / Trips / Falls 8. Manual Handling 9. Working at Height

	DEVELOPING PROCESS EVENT SCENARIOS GUIDELINE DOC NO: IS-HSE-GDL-310 REV. 01
---	--

33.Nucleonics 34.Pyrophoric		
DIVING 1. Water Jetting 2. HP Grit Blasting 3. Reduced Visibility 4. Cold Cutting 5. Oxy Arc Burning 6. O2 Build Up 7. Sharp edges / points 8. Hydraulic Tools 9. Excavating / Dredging 10.Air Bags / Air Lifting 11.Pressure / Vacuum 12.Collapsing / Overturning Structure 13.Entrapment 14.Permitted Umbilical Lengths 15.Extended Umbilical 16.Mid Water Swims 17.Falling Debris 18.Buoyancy 19.Thrusters 20.Habitat Welding 21.Strong Tidal Flow 22.Restricted Access 23.Drilling Muds 24.Complexity of Structure 25.ROV operations	DP 1. Reference Systems 2. Artemis Interface 3. Spotting of Taut Wire 4. Blow on / Blow Off 5. Within Anchor Pattern 6. Shallow Water Ops 7. Loss of Position 8. Thruster Wash	ROV 1. Propellers 2. Electric Power Source 3. Comms with Diver

	DEVELOPING PROCESS EVENT SCENARIOS GUIDELINE DOC NO: IS-HSE-GDL-310 REV. 01
---	--

APPENDIX D - REVISION HISTORY LOG

TABLE 3: REVISION HISTORY LOG

Revision Number: 01

Date: 18/09/2023

Item Revised	Revision Description	Page No.
All	Document simplified as part of QMR initiative.	All
Remarks:		

APPENDIX E - LIST OF ABBREVIATIONS

TABLE 4: LIST OF ABBREVIATIONS

Abbreviation	Description
CME	Critical Mitigation Element
DSV	Dive Support Vessel
HAZOP	Hazard and Operability Study
IPL	Independent Protection Layers
KME	Key Mitigation Element
LOPA	Layers of Protection Analysis
PHR	Process Hazard Review
PII	Permanent Incapacitating Injury